XD–TAU

Binary–Spherical Organogram Architecture

A proposed formal specification for linked hierarchy, spherical navigation and self-referential binary state.

One evolving state. Several linked representations.
Binary addresses connect a flat organogram, spherical regions, geometric fields and packed memory through explicit mappings.

INCLUDING AN OPTIONAL ADDENDUM

Shark--Geyser, barycentric log-polar encoding and the τ-axis spiral-screw extension.

Tom Klootwijk

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DOCUMENT GUIDE

Reading this specification

This document presents the preceding XD--TAU formalization as a standalone architectural specification. It retains the intended application: navigating populations or organizations over a sphere through a flat organogram. The new source is kept in a separate, optional addendum rather than made a dependency of the core. [S1, p. 12; S3, pp. 1–6; F0]

> **Status of the mathematics.** The core equations are the proposed definitions from the preceding formalization, not a claim that every equation originated in the supplied notes. The addendum distinguishes transcribed source expressions from compatibility checks and proposed integration rules. Neither part is presented as a benchmarked implementation or a demonstrated physical device.

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Source commitments The user's stated design vocabulary and intended use, identified by source and page.

Proposed definitions Explicit mathematical choices that make the architecture readable and testable. They remain design choices where the sources do not determine a unique rule.

Direct consequences Statements obtained from the displayed definitions, such as address uniqueness or the symmetry of the modulation matrix.

Open items Missing operational rules, measurements or physical models. They are identified rather than silently supplied.

Source keys S1--S3 refer to the supplied PDF exports. F0 is the preceding formalization in this conversation. Page references refer to the original PDFs, not this document.

1 The paradigm and its shared state

> XD--TAU is a time-stepped information architecture in which a flat hierarchy, a spherical spatial index, a log-polar lookup table, geometric fields and packed binary state are linked representations of one evolving system.

The source commitments are the organic organogram and L-system, binary branching, the first sphere-halving τ-step, OTAN2, the τ-pinion, signed-distance fields, two 45-degree blend operators, double Hadamard products, one-bit sampling and self-reference. [S1, pp. 1–4, 8, 11; S2, p. 1]

Architectural tuple

The proposed architecture is

$$\mathfrak{x} = \left(\Delta,\ S_R^2,\ \delta,\ \mathcal{L},\ \mathcal{G},\ \mathcal{B},\ \mathcal{U}\right). \tag{1.1}$$

Component	Role in the specification
Δ	Ordered sequence of observations or computational updates.
S_R^2	Spherical surface on which spatial entities are represented.
δ	Binary hierarchy for partitioning, addressing and navigation.
$\mathcal{L}$	Lookup relationships linking binary addresses, the flat organogram and spherical regions or coordinates.
$\mathcal{G}$	Geometric-field construction, including primitives, blends and Klein-bottle-derived geometry.
$\mathcal{B}$	Packed binary representation of data and operator descriptions.
$\mathcal{U}$	Update evaluator through which the current state and its stored rules generate the next state.

The defining relationship

$$\begin{aligned} \text{binary address} &\longleftrightarrow \text{hierarchical location} \\ &\longleftrightarrow \text{spherical region} \\ &\longleftrightarrow \text{stored state.} \end{aligned} \tag{1.2}$$

These arrows denote declared lookup relationships. They do not assert that a tree, sphere, cone, Klein bottle and bitstream are geometrically identical, or that every representation alone contains all the information in the others.

Organizational and spatial meaning A shared entity identity coordinates the views. A geographic partition is not automatically a reporting relationship. A change of position can leave an organizational parent unchanged, and a reorganization can leave geographic position unchanged.

Intended use The user's explicit application is rapid movement or navigation over a spherical surface from a flat population or organization chart. No dependence on the user's online profiles, kernels or models is introduced. [S1, p. 12; S2, p. 3]

2 Time and the first hemispherical cut

The source uses τ for a full turn, a step and a pinion. The proposed notation separates those roles without changing the design vocabulary. [S1, pp. 3–4, 8; S2, p. 1]

$$\tau_\circ = O = 2\pi, \qquad \mathcal{T}_n = \text{step } n, \qquad \omega_n = \tau_{\text{pinion},n}. \tag{2.1}$$

The one-dimensional arrow of time is the ordering

$$X = \Delta : \; n \longrightarrow n+1, \qquad T_n = \log\!\left(1 + \frac{t_n}{t_*}\right), \quad t_n \geq 0, \quad t_* > 0. \tag{2.2}$$

T_n is a proposed dimensionless log-time coordinate. This is an ordering of computation or observation, not by itself a physical theory of time.

First cut: full sphere to two hemispherical regions

To make ‘‘the first step halves the sphere’’ literal, begin with the full sphere:

$$S^2_R = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = R^2\}, \qquad R > 0. \tag{2.3}$$

Use the plane $y = 0$, matching the orientation of the source diagrams, and assign the boundary to the second region:

$$b_1(p) = \begin{cases} 0, & y < 0, \\ 1, & y \geq 0, \end{cases} \qquad \Pi_0 = S^2_R \cap \{y < 0\}, \qquad \Pi_1 = S^2_R \cap \{y \geq 0\}. \tag{2.4}$$

Thus $\mathcal{T}_1 : S^2_R \to \{\Pi_0, \Pi_1\} \to \{0, 1\}$. The source illustrations show a dome; using the full sphere as the pre-cut domain is an explicit convention of this specification. [S2, pp. 7–9]

The apex is $A = (0, 0, R)$. A cone with this apex and an equatorial circular base is bisected by the same plane; its axial section is triangular. This links the apex, cone, circle and side profile without identifying them as one shape.

Recursive refinement

For a binary word w, start with $C_\varnothing = S^2_R$. Choose a separator $g_w(p) = \mathbf{n}_w \cdot p - c_w$, with $\|\mathbf{n}_w\| = 1$, and set

$$C_{w0} = C_w \cap \{g_w < 0\}, \qquad C_{w1} = C_w \cap \{g_w \geq 0\}. \tag{2.5}$$

Equal-area refinement requires the additional rule $\mu(C_{w0}) = \mu(C_{w1}) = \mu(C_w)/2$, where μ is spherical area. Equal population counts require a different balancing measure. Time step and tree depth are distinct: one update may refine one region, several regions or none.

3 The binary organogram and its addresses

A node at depth d has an address $w = b_1 \cdots b_d$. Deleting its final bit gives the parent; appending 0 or 1 gives its children. A minimal L-system production is

$$N(w) \longrightarrow N(w0)\,N(w1). \tag{3.1}$$

This production supplies branching structure. Positions, lengths, field parameters and appearance belong to a separate geometric interpretation. The source explicitly combines an L-system with a log-polar tree mapping. [S1, p. 2]

Proposed log-polar layout

Let

$$j(w) = \sum_{i=1}^{d} b_i 2^{d-i}, \qquad I_w = \left[\frac{2\pi j(w)}{2^d}, \frac{2\pi(j(w)+1)}{2^d}\right). \tag{3.2}$$

Assign the displayed angular center and radial scale

$$\theta_w = \frac{2\pi\left(j(w)+\frac{1}{2}\right)}{2^d}, \qquad r_d = r_0 a^d, \quad r_0 > 0, \quad a > 1. \tag{3.3}$$

Then

$$\rho_d = \log(r_d/r_0) = d\log a, \qquad \mathcal{L}_{\text{org}}(w) = (\rho_d, \theta_w). \tag{3.4}$$

Successive hierarchy levels are equally spaced in log radius, while branching subdivides each angular sector. The displayed radius is a hierarchy coordinate, not automatically geographic distance.

Logical node address and entity identity

A collision-free logical node number is

$$\text{addr}(w) = 2^{|w|} - 1 + j(w). \tag{3.5}$$

At depth d, these numbers occupy $[2^d - 1,\ 2^{d+1} - 2]$; depth ranges do not overlap. Sparse trees can use a compact node table rather than allocating every possible address.

An entity record keeps organizational and geographic meanings separate:

$$e = (\text{ID},\ \text{organizational parent},\ \text{spatial address},\ \text{payload}). \tag{3.6}$$

Several entities may share a spatial cell. Identity is therefore not replaced by a cell number, an occupancy bit or a screen position.

> **Binary tree versus binary search tree.** The geometric construction defines a binary partition tree. A search-tree implementation must additionally maintain an ordering condition on the keys in each left and right subtree. Binary branching alone does not provide that condition. The geometric tree and any ordered search structure must have a declared relationship.

4 Spherical lookup and coordinate domains

The lookup layer connects tree addresses to the regions selected by the geometric separators. A displayed organogram angle and a physical spherical angle need not coincide. This separation prevents an attractive flat layout from silently changing the meaning of geographic position.

Proposed hemispherical parameterization

For the cut $y = 0$, let $h \in \{0, 1\}$ identify a hemisphere and let $s_h = 2h - 1$. Use

$$p = R\big(\sin\varphi\cos\psi,\ s_h\cos\varphi,\ \sin\varphi\sin\psi\big), \tag{4.1}$$

with

$$0 \le \varphi \le \frac{\pi}{2}, \qquad 0 \le \psi < 2\pi. \tag{4.2}$$

The equatorial boundary belongs only to $h = 1$, in accordance with Section 2. The symbols φ and ψ denote angles here; an independent branch-scaling coefficient must have a different name.

The projected planar radius is

$$r = R\sin\varphi = \sqrt{x^2 + z^2}, \qquad 0 \le r \le R. \tag{4.3}$$

A finite lookup implementation can use the proposed regularized log coordinate

$$\rho = \log\left(1 + \frac{r}{r_*}\right), \qquad r_* > 0, \tag{4.4}$$

so its permitted range is

$$0 \le \rho \le \log\left(1 + \frac{R}{r_*}\right). \tag{4.5}$$

This explicitly differs from the unregularized $\log r$ expression in the source. It keeps the radial coordinate finite at $r = 0$; it does not remove the angular ambiguity at a pole. [S1, pp. 2, 4; F0]

What the lookup must retain

The table must identify the relevant tree word or cell, hemisphere, coordinate convention and entity records. A region representative may be adequate for an aggregate view; exact entity positions require their own precision or residual information. The same lookup identity is used when switching between the organogram and the sphere.

> **Two radii, two meanings.** The organogram's $r_d = r_0 a^d$ expresses hierarchy depth. The spherical chart's $r = \sqrt{x^2 + z^2}$ expresses projected spatial radius. A declared lookup connects the representations; equating their radii is not a default rule.

Poles and seams Each pole requires one designated angular address, since every angle represents the same point at $r = 0$. The hemispherical seam needs explicit cross-seam neighbors and a single ownership convention. A display phase may change while the underlying entity position remains fixed.

CORE SPECIFICATION / CONTINUED

OTAN2 and the τ-pinion

Turn-valued angular addressing

A precise candidate meaning for the source's named operator is

$$\mathrm{OTAN2}(z, x; T_n) = \mathrm{frac}\left(\frac{\mathrm{atan2}(z, x)}{2\pi} + \frac{\omega_n}{2\pi}\right). \tag{4.6}$$

The output $q \in [0, 1)$ is measured in turns, so one unit is a full revolution. A proposed phase law is

$$\omega_n = [\omega_0 + \nu T_n]_{2\pi}, \tag{4.7}$$

where $[\cdot]_{2\pi}$ means reduction modulo 2π and ν is phase advance per unit of log-time. This gives OTAN2 and the pinion explicit roles without prescribing whether the angle is computed, approximated or retrieved from a table. [S1, pp. 3–4, 8; F0]

Why the time factor is explicit

The source's earlier AI-generated expression places a common $\log T$ factor inside both arguments of an inverse tangent. For any positive factor c,

$$\mathrm{atan2}(cz, cx) = \mathrm{atan2}(z, x). \tag{4.8}$$

The scaled point is on the same ray. Thus the common positive scaling does not generate a continuous time-dependent angular advance. The phase law above is an explicit proposed replacement, not a restatement of that source equation. [S1, p. 4]

Return map to the sphere

Given a legal ρ, the turn coordinate q and the hemisphere h, define

$$r = r_*(e^{\rho} - 1), \qquad \psi = [2\pi q - \omega_n]_{2\pi}. \tag{4.9}$$

The return map is

$$\boxed{p = \left(r\cos\psi,\ s_h\sqrt{R^2 - r^2},\ r\sin\psi\right), \quad 0 \le r \le R.} \tag{4.10}$$

It satisfies the sphere constraint directly:

$$\|p\|^2 = r^2\cos^2\psi + (R^2 - r^2) + r^2\sin^2\psi = R^2. \tag{4.11}$$

At a pole, use the designated angular address; at $r = R$, use the assigned seam owner. The inverse relationship is between valid coordinate representatives, not between distinct pole angles that already denote the same point.

> **Coordinate motion is not physical motion.** The pinion can rotate the view or its angular address convention without moving an entity. Physical movement, or a recorded change in position, requires a separate event or motion rule.

5 Signed-distance geometry and blends

The source explicitly requires the signed-distance field to remain part of the accumulated operator architecture. The proposed specification separates an implicit scalar field, its sign and its distance to a boundary. [S1, pp. 3, 6, 11]

A true distance layer

For a region $\Omega \subset \mathbb{R}^3$ with a nonempty boundary, use negative values inside:

$$d_\Omega(p) = \begin{cases} -\inf_{q\in\partial\Omega} \|p-q\|, & p \in \Omega, \\ +\inf_{q\in\partial\Omega} \|p-q\|, & p \notin \Omega. \end{cases} \tag{5.1}$$

The information flow is

$$\Phi \longrightarrow \Omega = \{p : \Phi(p) \le 0\} \longrightarrow d_\Omega. \tag{5.2}$$

The earlier source expression $\operatorname{sgn}(\Phi)\|\nabla\Phi\|$ is not a general distance reconstruction. For $\Phi(x, y, z) = x$, it returns $\operatorname{sgn}(x)$, whereas the signed distance to $x = 0$ is x. This explicit counterexample is why the distance layer is kept separate. [S1, p. 11]

Candidate semantics for the two operators

The source requests a 45-degree encircled blend and a non-rotated counterpart but does not uniquely determine dimensionally consistent operators. The following are the candidate definitions used in the preceding formalization. [S1, pp. 3–5; F0]

Keep the angle $\alpha = \pi/4$ separate from a smoothing length $h_b > 0$. For two scalar field values a, b, define

$$a \boxplus b = \min(a, b). \tag{5.3}$$

For the encircled blend, rotate the *field-value* coordinates by 45°:

$$u = \frac{a+b}{\sqrt{2}}, \qquad v = \frac{b-a}{\sqrt{2}}, \qquad a \oplus_{h_b} b = \frac{u - \sqrt{v^2 + h_b^2}}{\sqrt{2}}. \tag{5.4}$$

As $h_b \to 0$, the encircled blend approaches $\min(a, b)$ because

$$\frac{a + b - |b-a|}{2} = \min(a, b). \tag{5.5}$$

A 45-degree rotation in the (a, b) value plane is not automatically a 45-degree cutting plane in physical space. Such a physical cut still requires a geometric separator. Likewise, $O = 2\pi$ can govern phase without multiplying every distance-valued expression by 2π.

CORE SPECIFICATION / CONTINUED

Klein-bottle geometry and the accumulated field

A declared interpretation of the Klein-bottle component

For this specification, use a thickened image of a chosen Klein-bottle immersion. Let

$$\iota : K \longrightarrow \mathbb{R}^3, \qquad \Omega_{K,\epsilon} = \{p : \operatorname{dist}(p, \iota(K)) \leq \epsilon\}, \quad \epsilon > 0, \tag{5.6}$$

and define

$$f_K = d_{\Omega_{K,\epsilon}}. \tag{5.7}$$

This is a proposed way to assign a solid/void region to the Klein-bottle-derived component. It does not identify the abstract surface with its self-intersecting image. A specific immersion, scale, thickness and numerical representation remain implementation choices. [S1, pp. 3, 6, 8–9; F0]

The construction does not, by itself, prove that intersections create useful chambers, preserve spherical adjacency or add storage capacity. Those are separate geometric or computational properties to construct and test.

Accumulating the fields

Let f_{pyr}, f_{cone} and f_{sphere} denote declared three-dimensional primitive fields with compatible length units. A triangular pyramid side profile requires an explicit three-dimensional realization, such as an extrusion; a two-dimensional profile is not silently treated as a three-dimensional distance field.

Let a lookup select a dimensionless modulation value from the matrix in Section 6:

$$m_n(p) = M_{i_n(p),j_n(p)}. \tag{5.8}$$

Then retain the proposed accumulated expression

$$\boxed{\Phi_n(p) = \left(f_{\text{pyr}} \boxplus f_{\text{cone}}\right) \oplus_{h_b} \left(f_{\text{sphere}} \boxplus [m_n(p) f_K]\right).} \tag{5.9}$$

All fields on the right are evaluated at p. The final geometry and its distance are

$$\Omega_n = \{p : \Phi_n(p) \leq 0\}, \qquad d_n = d_{\Omega_n}. \tag{5.10}$$

This order matters: modulation and blending generate an implicit field, but are not assumed to preserve exact signed distances. In particular, a spatially changing sign mask may introduce discontinuities. The distance is to the boundary of the resulting declared region, not merely to a convenient zero sample of an intermediate expression.

Metric and domain. The fields in this section use Euclidean distance in $\mathbb{R}^3$. Distances used for navigation along S^2_R are surface distances and belong to a different metric. Geometry generation and spherical routing must not silently exchange these meanings of distance.

6 Double Hadamard products and packing

Retain the source's double elementwise product, with the explicit proposed input domain $A, B \in \{-1, +1\}^{N \times N}$:

$$M = (A \odot B) \odot (A^{\mathsf{T}} \odot B^{\mathsf{T}}). \tag{6.1}$$

Here $\odot$ means elementwise multiplication, not a matrix product or a Walsh--Hadamard transform. [S1, p. 8]

What follows from the expression

Put $C = A \odot B$. Then

$$M = C \odot C^{\mathsf{T}}, \quad M_{ij} = C_{ij}C_{ji}, \quad M_{ij} = M_{ji}, \quad M_{ii} = 1. \tag{6.2}$$

Thus M is a symmetric, sign-valued modulation matrix. It is not automatically an orthogonal transform, sorting method or error-correcting code. For the explicit case $A = B$, every entry of C and M equals +1; the construction then removes all variation.

Bitwise form

Encode signs by $s = (-1)^b$, $b \in \{0, 1\}$. Since

$$(-1)^a(-1)^b = (-1)^{a \,\mathrm{XOR}\, b}, \tag{6.3}$$

the bit encoding of the modulation is

$$m_{ij}^{\mathrm{bit}} = a_{ij} \,\mathrm{XOR}\, b_{ij} \,\mathrm{XOR}\, a_{ji} \,\mathrm{XOR}\, b_{ji}. \tag{6.4}$$

The distinct notation XOR avoids confusing a binary logical operation with the encircled geometric blend $\oplus_{h_b}$.

Storage is a separate contract

''Doubly packed'' needs a concrete storage convention. The proposal is to pack the two input bit planes separately and derive the modulation mask when needed. For a packed array W of 64-bit unsigned words, logical bit i is

$$\mathrm{bit}(W, i) = \left(W_{\lfloor i/64 \rfloor} \gg (i \bmod 64)\right) \,\&\, 1. \tag{6.5}$$

$\gg$ is a right shift and & is bitwise AND. The collision-free tree address from Section 3 can identify logical node slots, but record layout, offsets and bounds still need to be specified. An entity record will generally occupy multiple bits.

> **Packing is not proof of compression.** The matrix operation, the binary sign encoding, the memory layout and the field sampling are different parts of the architecture. Their useful properties must be established separately rather than inferred from the word ''Hadamard'' or ''one-bit.''

7 The self-referential state transition

The source's self-reference requirement is that stored bits select the operations used to evaluate a field, and that the evaluation participates in rewriting the stored state. [S1, pp. 11–12]

Let $B_n \in \{0,1\}^*$ be the finite encoded state. Its decoded content includes the entity records, tree, lookup configuration, primitive parameters, Hadamard inputs, phase parameters and update-rule description. With external event e_n, define

$$\boxed{\begin{aligned} s_n &= \mathrm{Decode}(B_n), \\ p_{n,i} &= \mathcal{L}(s_n, i), \\ q_{n,i} &= \mathbf{1}\left[d_{s_n}(p_{n,i}) + J_{n,i} \le 0\right], \\ s_{n+1} &= \mathcal{U}(s_n, e_n, q_n), \\ B_{n+1} &= \mathrm{Encode}(s_{n+1}). \end{aligned}} \tag{7.1}$$

Here $J_{n,i} \in [-\epsilon_J, \epsilon_J]$ is bounded jitter, and a sample bit of 1 denotes the sampled interior. $\mathcal{U}$ evaluates the update description in the current state; it must specify how data, operator descriptions and samples affect the next state.

> **Central self-reference.** The current encoded description governs geometric evaluation, and that result contributes to the next encoded description. This is a feedback architecture, not an assertion that a geometric shape is literally a computer program without an encoding and evaluator.

Record preservation The update must retain entity identities and unrelated payloads. A single occupancy sample must not overwrite an entire entity record. The reading of the current state and writing of the next state also need a declared evaluation order.

Reproducibility Repeatable jitter requires a fixed generation rule and seed or generator state. Intentional state changes must remain distinguishable from visual dithering.

Binary storage versus one-bit samples All finite records can be encoded as sequences of bits. A single field sample, however, records one classification, not an arbitrary distance. For example,

$$\mathbf{1}[-1 \le 0] = \mathbf{1}[-100 \le 0] = 1. \tag{7.2}$$

Recovering geometry therefore requires sample locations, model parameters, additional samples or additional numerical information.

Stationarity is a separate question For fixed external conditions, a stationary state would satisfy $B_* = \mathcal{F}(B_*)$. Its existence or stability, and any self-reproduction or error-correction property, depend on the actual update rule. The source does not determine a unique $\mathcal{U}$; these equations specify an architecture, not one fully determined executable algorithm.

8 Application, correctness and open choices

The intended workflow is to select an entity or subtree in the flat organogram, resolve its binary address, retrieve its spherical representation, navigate or update, and refresh the linked views. Increasing depth can request finer spatial resolution; selecting a subtree can identify a group. [S1, pp. 12–13; F0]

A position update changes spatial information while preserving entity identity. A reporting-line update changes organizational information without requiring geographic movement. The common lookup relationship coordinates the displays.

Minimum correctness requirements

Requirement	What must hold
Identity	The same entity remains identifiable across representations and updates; shared cell occupancy does not merge records.
Spatial accuracy	A decoded approximation $\widehat{p}$ satisfies a declared surface-distance tolerance $d_{S^2_R}(p, \widehat{p}) \leq \varepsilon_{\text{space}}$.
Addressing	Valid cells have unambiguous addresses; poles, seam ownership and cross-seam neighbors are explicitly handled.
State integrity	Operator updates, payload updates and rendering jitter have distinct, reproducible roles.

Performance is a testable hypothesis

Traversing a depth-d address makes d branch decisions. Logarithmic depth in the number of represented items requires an appropriate balancing condition. Lookup tables can replace repeated calculations with retrieval, but their construction, storage and update costs remain part of the system.

The source exports assert instantaneous movement, automatic self-healing and major computational savings without supplying measurements that establish those results. These statements remain hypotheses, not guaranteed properties of this specification. [S1, pp. 12–13]

Choices needed for an implementation

The remaining choices include the recursive separator and balancing rule; the LUT resolution and neighbor scheme; the exact primitive fields and Klein-bottle immersion; the field-sampling and packing formats; and the update evaluator $\mathcal{U}$. Each choice should have a declared input domain, precision and failure behavior.

> **Consolidated statement.** XD--TAU is Tom Klootwijk's proposed binary--spherical organogram architecture: a time-ordered, self-referential system in which hierarchy, spatial representation, geometric evaluation and update descriptions are coordinated through a common addressable state. Its purpose is navigation and restructuring of populations or organizations across linked flat and spherical views.

A Optional Shark–Geyser addendum

A.1 / SOURCE SEQUENCE AND SUPPLIED EXPRESSIONS

The new source adds the ‘‘shark sawtooth,’’ a two-stage geyser (‘‘samen’’), barycentric log-polar encoding and a Len--Θ radial output. The order and terminology below follow that document. This addendum is optional and is not a prerequisite for the core architecture. [S3, pp. 1–3]

> **Source status.** Expressions are transcribed with disambiguated local symbols; they do not redefine core variables. Missing functions remain unspecified. The integration contract on page 15 is a new proposal, not a recovered implementation.

1. Shark sawtooth

The source supplies a periodic folded profile:

$$f_p(x) = \left|(x \bmod p) - \frac{p}{2}\right|, \qquad p > 0. \tag{A.1}$$

Its stated role is to introduce a serrated or triangular modulation before the radial stages. The source does not specify a complete two-dimensional coordinate map or how this profile couples to every later variable. [S3, p. 2]

2. Geyser: chamber and eruption, together

Stage 2A, the chamber, is described as nonlinear accumulation or compression toward a node; no equation is supplied. **Stage 2B, the eruption,** supplies

$$r' = r_0 \exp\big(\alpha_g (r - r_c)\big). \tag{A.2}$$

The source uses α here; α_g distinguishes it from angular intensity below. ‘‘Thermal’’ and ‘‘hydrothermal’’ describe the source’s analogy, not a supplied pressure, temperature or conservation model. [S3, p. 2]

3. Barycentric log-polar encoding

The source specifies three weights and a logarithmic radial transformation:

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \qquad \rho_{\text{raw}} = \ln(r). \tag{A.3}$$

It describes a reference triangle but does not provide its vertices or the full mapping into it. The raw logarithm is retained here as source notation; it is not silently replaced by the core’s regularized coordinate. A dimensional radius needs a declared reference scale before use inside a logarithm. [S3, p. 2]

4. Len–Θ radial output

Len is described as radial magnitude derived from logarithmic distance and sawtooth modulation; Θ is the accumulated angular phase. The source does not give a unique equation for Len or the complete barycentric-to-output map. In particular, this addendum does not assume Len = ρ merely because both are radial quantities. [S3, p. 3]

ADDENDUM A / CONTINUED

The τ-axis and the 2D/3D screw

A.2 / SOURCE EXTENSION

The user adds an angular shockwave along an eigenvector axis τ, then a simultaneous 2D/3D spiral screw with repeating sinusoidal segments, dissipation, diffusion and a ''VantaBlack-style'' material analogy. [S3, pp. 3, 5]

Eigenvector direction and angular phase

The source supplies $v_\tau = (x_\tau, y_\tau)^\mathsf{T}$ and $\tau = \operatorname{atan2}(y_\tau, x_\tau)$. Denote that angle by β here to avoid confusing it with a full turn, an update step or the pinion phase. The underlying matrix and a rule for choosing its eigenvector are not specified. [S3, p. 4]

The supplied phase expression is

$$\Theta(\theta) = \theta + \alpha_\theta \tanh\left(\frac{\theta - \beta}{\sigma}\right) \sin(2\pi\,\mathrm{Len}), \qquad \sigma > 0. \tag{A.4}$$

The source uses α for this intensity as well; α_θ separates it from α_g. A dimensionless Len convention, an angular branch and a seam rule would have to be declared. The source calls this a shockwave; for fixed Len and finite $\sigma > 0$, the expression is smooth in θ, not a literal jump. [S3, p. 4]

Spiral-screw relation and repeating segments

The source gives the axial pitch relation and modulation envelope

$$z = k\Theta, \qquad I(r) = \sin\big(\omega_s \ln(r)\big). \tag{A.5}$$

Here ω_s denotes the source's segmentation frequency ω, not the core pinion phase ω_n. The logarithm requires $r > 0$ and a declared normalization. The screw relation requires an unwrapped phase or a retained turn count: reducing Θ modulo 2π alone loses the axial displacement $2\pi k$ per turn. [S3, pp. 5–6]

The source's ''infinitely repeating'' language is retained as an idealized scale-repetition description. It does not establish infinite energy or a finite-resolution implementation. The effect of a negative envelope on radius, pitch or visibility is not specified.

Dissipation, diffusion and the material analogy

The source proposes counter-Coriolis deflection and an exponential decay factor

$$\exp(-\mu t). \tag{A.6}$$

For $t \geq 0$, the factor is non-growing when $\mu \geq 0$, with strict decay for $\mu > 0$. No complete spatial diffusion equation, calibrated material response, absorption measurement or runnable simulation is included in the PDF. Its ''VantaBlack-style'' description therefore remains an analogy, not an established material property of the architecture. [S3, pp. 5–6]

ADDENDUM A / INTEGRATION CONTRACT

Compatibility without interference

A.3 / PROPOSED INTEGRATION AND DIRECT CHECKS

> **Compatibility decision.** Include the source as a detachable view or simulation extension, disabled by default. Do not replace the canonical address map, spherical position records, distance definition or state-transition rule with its coordinate warps.

Let $V_n = \mathrm{View}(B_n)$ be a derived view and let η contain optional module settings. The proposed boundary is

$$\widetilde{V}_n = \mathcal{G}_\eta(V_n), \qquad \mathcal{G}_{\mathrm{off}} = \mathrm{id}, \qquad B_{n+1} = \mathcal{F}(B_n, e_n). \tag{A.7}$$

η is kept separate from the canonical state, and $\widetilde{V}_n$ is not fed into $\mathcal{F}$. Every rendered item retains its entity ID and original coordinates. Hit-testing resolves those retained identities rather than assuming the warped coordinates have an inverse. This separation leaves the core equations unchanged even when a display deformation folds or overlaps.

Checks on the source expressions

Periodic folding is not unique addressing. The supplied profile satisfies $f_p(x + p) = f_p(x)$ and $f_p(p/4) = f_p(3p/4)$. Its triangular corners do not make it injective. Used alone as a key, it would merge distinct positions. [S3, p. 2]

Log coordinates do not confer automatic invariance. For $a > 0$, $\ln(ar) = \ln(r) + \ln(a)$. Scaling becomes a shift in the raw log coordinate, not an unchanged value. Triangle containment also needs nonnegative barycentric weights and a declared nondegenerate reference triangle; a sum of one alone is insufficient. [S3, p. 2]

Ranges and seams remain open. Exponential expansion may exceed the core radius range. The angular expression needs a periodic seam convention, the screw needs a turn count, and $\ln r$ needs a positive lower cutoff for a finite implementation. None is supplied as a complete rule. [S3, pp. 2, 4–6]

Deflection is not a damping proof. If the source's counter-Coriolis term is represented by $\pm 2\boldsymbol{\Omega} \times v$, then

$$v \cdot (\pm 2\boldsymbol{\Omega} \times v) = 0. \tag{A.8}$$

Changing that sign alone does not supply an energy-loss term. The separate exponential factor is a proposed decay model, not evidence of perfect absorption or diffusion. [S3, pp. 5–6]

Result: the extension is compatible under the view-only boundary above. Promoting it into routing, SDF evaluation or feedback would be a separate architectural change requiring explicit maps, distance reconstruction, bounds and tests. The source does not yet establish that stronger form of compatibility.

PROVENANCE

Sources and edition notes

The source PDFs are conversation exports containing both the user's instructions and AI-generated responses. They establish the design vocabulary and the sequence of requests; the generated explanations are not independent experimental evidence.

S1 / Organic organogram and log-polar architecture

Supplied PDF, 14 pages. Opening user request: ''organic oragnogram where binary search tree is mapped to a log encoded polar LUT L-system'' followed by the geometric primitives, signed-distance field and one-bit jitter.

Pages 2--5 introduce the tree, log-polar mapping, L-system, OTAN2 and blend expressions. Pages 6 and 8--11 develop the signed-distance, Klein-bottle, Hadamard and self-reference material. Page 12 states the intended population/organization navigation application; page 13 supplies the unmeasured performance claims.

S2 / XD–TAU and the first hemispherical cut

Supplied PDF, 9 pages. Opening user request: ''XD TAU where X is the 1D arrow of time Delta'' and a binary search tree along a hemisphere, with the first step halving the sphere.

Page 1 establishes the naming and initial cut. Page 3 excludes the user's kernels and models from the requested formalization. The diagrams on pages 7--8 and their explanation on page 9 use the $y = 0$ cutting plane and an apex/root interpretation.

S3 / `geiser-haailen-samen.pdf`

Supplied PDF, 6 pages. Page 1 gives the user's shark-sawtooth / two-stage geyser / barycentric log-polar request. Pages 2--3 supply the stage sequence and Len--Θ framing. Page 4 supplies the axis and angular formula. Pages 5--6 supply the 2D/3D screw, sinusoidal envelope and dissipation/material analogy.

The initial household-appliance interpretation on page 1 is not part of the geometric addendum. The file says that simulations were generated or updated, but does not include executable simulation code or results establishing the stated physical behavior.

F0 / Preceding formalization in this conversation

The response titled *XD--TAU: Binary--Spherical Organogram Architecture* is the basis of the main specification. Its proposed equations are restated here with typography and symbol separation for readability. They are not attributed wholesale to the original user notes.

> **Edition 1.1.** Retains the core architecture, adds the optional three-page S3 addendum, and separates the geometric blend, logical XOR, smoothing length, segmentation frequency and eigenvector-axis angle. No update algorithm, benchmark result or physical material model has been silently added. Attribution and identifying details are reproduced as supplied by Tom Klootwijk.